

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Data Warehousing and Data Mining COURSE CODE: CSA1603

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COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

Annexure A
MCQ TEST

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I P.GAYATHRI/192425277 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ONLINE TRANSACTION PROCESSING (OLTP)



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INTRODUCTION

- **OLTP (Online Transaction Processing)** is a database system that manages **real-time transactions**.
- It is designed to **process a large number of short and fast transactions**.
- OLTP ensures **accurate, secure, and reliable** data processing.
- It supports **multiple users** accessing the database simultaneously.
- It is widely used in **banking, e-commerce, ATM, ticket booking, and online payment systems**.
- The main objective of OLTP is to **provide quick response time while maintaining data integrity**.

FEATURES OF OLTP

- **Real-Time Processing:** Processes transactions immediately as they occur.
- **Fast Response Time:** Completes transactions within seconds.
- **High Concurrency:** Supports many users accessing the database simultaneously.
- **Data Integrity:** Maintains accurate and consistent data.
- **ACID Properties:** Ensures transactions are reliable, secure, and error-free.
- **High Availability:** Provides continuous system operation with minimal downtime.



WORKING OF OLTP

- **User:** Initiates a transaction (e.g., payment, booking, or withdrawal).
- **Application:** Receives and validates the user's request.
- **Database:** Stores and updates the transaction data securely.
- **Transaction Processing:** Executes the transaction while ensuring accuracy and consistency.
- **Response:** Returns a confirmation or error message to the user instantly.

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APPLICATIONS OF OLTP

- **Online Banking** – Manages deposits, withdrawals, and fund transfers in real time.
- **ATM Transactions** – Processes cash withdrawals, balance inquiries, and mini statements instantly.
- **E-Commerce** – Handles online orders, payments, and inventory updates.
- **Ticket Booking Systems** – Enables railway, airline, and movie ticket reservations.
- **Hospital Management** – Stores patient records, billing, and appointment details.
- **Online Payment Systems** – Supports UPI, credit/debit card, and digital wallet transactions.

ADVANTAGES & DISADVANTAGES OF OLTP

Advantages

- **Fast Transaction Processing** – Executes transactions quickly in real time.
- **High Data Accuracy** – Ensures correct and consistent data.
- **Supports Multiple Users** – Allows many users to access the system simultaneously.
- **Secure and Reliable** – Protects data and maintains transaction integrity.

Disadvantages

- **Limited Data Analysis** – Not suitable for complex reporting or analytics.
- **Requires Regular Backup** – Frequent backups are needed to prevent data loss.
- **High Maintenance Cost** – Requires powerful hardware and continuous system maintenance.
- **Performance Depends on Database** – Heavy traffic can slow down the system.

CONCLUSION

- **OLTP (Online Transaction Processing)** is a reliable system for managing **real-time transactions**.
- It provides **fast, accurate, and secure** transaction processing.
- OLTP supports **multiple users** while maintaining **data integrity and consistency**.
- It is widely used in **banking, e-commerce, ATM, ticket booking, and online payment systems**.
- Overall, OLTP plays a vital role in ensuring **efficient and reliable business operations**.



Thank You

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand